

CH 1 — Abstract Vector Spaces

Luke Lu • 2026-09-23

1.1 Vector Spaces

Ideas: $\mathbb{R}^n$ is useful. There are other objects that “behave” and “look” like $\mathbb{R}^n$

Examples:

1. $V = P_b(\mathbb{F}) = \{a_0 + a_1x + a_2x^2 + \dots + a_nx^n : a_i \in \mathbb{F}\}$

Polynomial degree $\leq n$:

Additive property

$$a_0 + a_1x + \dots + a_nx^n + b_0 + b_1x + \dots + b_nx^n = (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_n + b_n)x^n$$

Scalar Multiplicative property

$$c(a_0 + a_1x + \dots + a_nx^n) = ca_0 + ca_1x + \dots + ca_nx^n$$

2. $V = M_{m \times n}(\mathbb{F}) = \left\{ \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} : a_{ij} \in \mathbb{F} \right\}$

Same properties for matrices like the above

Observation:

$\mathbb{R}^4$	$P_3(\mathbb{R})$	$M_{2 \times 2}(\mathbb{R})$
$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$	$a + bx + cx^2 + dx^3$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Notice that the above not only “look alike”, they share similar operations (addition and scalar multiplication)

This is isomorphism

3. $V = F(\mathbb{R}, \mathbb{R}) = \{\text{functions}, f : \mathbb{R} \rightarrow \mathbb{R}\}$

We can add and scalar multiply

Question: Does $F(\mathbb{R}, \mathbb{R})$ is isomorphic to $\mathbb{R}^n$

ANS: No, the reason will be learned later in the course. The current explanation is that $F(\mathbb{R}, \mathbb{R})$ is an infinity-dimensioned vector space.

Definition – Vector Space

A vector space over $\mathbb{F}$ is a set V together with two operations $\begin{cases} \text{addition}(+) \\ \text{scalar multiplication}(\cdot) \end{cases}$ s.t.

$$\forall \vec{x}, \vec{y}, \vec{x} + \vec{y} \subseteq V$$

$$\forall c \in \mathbb{F}, c\vec{x} \subseteq V$$

Furthermore, the following conditions must hold for $\forall \vec{x}, \vec{y}, \vec{z} \in V, c, t \in \mathbb{F}$

1. $\vec{x} + \vec{y} = \vec{y} + \vec{x}$
2. $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$
3. $c(\vec{x} + \vec{y}) = c\vec{x} + c\vec{y}$
4. $c(t\vec{x}) = ct(\vec{x})$
5. $(s + t) \cdot \vec{x} = s \cdot \vec{x} + t \cdot \vec{x}$
6. $1 \cdot \vec{x} = \vec{x}$
7. $\exists \vec{0} \in V$ s.t. $\vec{x} + \vec{0} = \vec{x}$
8. $\forall \vec{x} \in V, \exists -\vec{x} \in V$ s.t. $\vec{x} + (-\vec{x}) = \vec{0}$

Warning – Zero Vector and Additive Inverse

Zero vector $\vec{0}$ is not the vector with every entry being 0. It is subject to change with its corresponding vector space

Additive Inverse $-\vec{x}$ are not the vector with every entry being the negation of its origin entry. Refer to the Axiom to determine the corresponding additive inverse

Both $\vec{0}$ and $-\vec{x}$ are **unique** in their corresponding vector space

Note:

1. The elements in V are called vectors
2. We omit the “.” in $s \cdot \vec{x}$ to $s\vec{x}$
3. We write $\vec{x} - \vec{y}$ instead of $\vec{x} + (-\vec{y})$
4. $\vec{0}$ is called the **zero vector**
5. $-\vec{x}$ is called the **additive inverse** of $\vec{x}$

Examples:

$$1. V = \mathbb{F}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{F} \right\}, V = M_{m \times n}(\mathbb{F}) = \left\{ \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} : a_{ij} \in \mathbb{F} \right\},$$

and $V = P_n(\mathbb{F}) = \{a_0 + a_1x + \dots + a_nx^n : a_i \in \mathbb{F}\}$ with addition and scalar multiplication be usual operations.

$$\text{Here, } \vec{0} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}, 0 + 0x + \dots + 0x^n$$

2. $V = F(\mathbb{R}, \mathbb{R}) = \{\text{functions}, f : \mathbb{R} \rightarrow \mathbb{R}\}$ with addition and scalar multiplication be usual operations.

Here, $\vec{0}$ = the zero function, i.e. $z(x) = 0$

3. $V = \mathbb{R}^2$

Define addition and scalar multiplication by

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_2 \\ x_2 + y_1 \end{bmatrix}$$

$$s \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} sx_2 \\ sx_1 \end{bmatrix}$$

This is NOT a vector space.

$$1\vec{x} \neq \vec{x} \implies \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \neq \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$$

$$\vec{x} + \vec{y} \neq \vec{y} + \vec{x}$$

4. $V = \mathbb{R}^2$

Define addition and scalar multiplication by

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 - 1 \\ x_2 + y_2 + 1 \end{bmatrix}$$

$$s \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} sx_1 + s - 1 \\ sx_2 - s + 1 \end{bmatrix}$$

Check axiom 7 and 8 as 1-6 are easy and straightforward

Axiom 7:

Note that in this case $\begin{bmatrix} 0 \\ \dots \\ 0 \end{bmatrix}$ is not the zero vector for this vector space!

We suppose $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$,

$$\text{then } \vec{x} + \vec{0} = \vec{x} \iff \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \iff \begin{bmatrix} x_1 + z_1 - 1 \\ x_2 + z_2 + 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \implies \vec{z} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \in V$$

Axiom 8:

We suppose $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$

$$\text{then } \vec{x} + \vec{z} = \vec{0} \iff \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \iff \begin{bmatrix} x_1 + z_1 - 1 \\ x_2 + z_2 + 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies \vec{z} = \begin{bmatrix} x_1 + 2 \\ x_2 - 2 \end{bmatrix}$$

$$-\vec{x} = \begin{bmatrix} -x_1 - 2 \\ -x_2 + 2 \end{bmatrix}$$

5. Proof uniqueness of $\vec{0}$

Let V be a vector space. Suppose $\vec{0}_1, \vec{0}_2 \in V$ satisfy

$$\vec{0}_1 + \vec{x} = \vec{x}, \vec{x} \in V$$

$$\vec{0}_2 + \vec{x} = \vec{x}, \vec{x} \in V$$

We want $\vec{0}_1 = \vec{0}_2$

Proof

$$\vec{0}_2 + \vec{x} + \vec{0}_1 = \vec{x} \implies \vec{0}_2 + \vec{0}_1 = \vec{0}_1$$

$$\vec{0}_1 + \vec{x} + \vec{0}_2 = \vec{x} \implies \vec{0}_1 + \vec{0}_2 = \vec{0}_2$$

$$\text{Together, } \vec{0}_1 + \vec{0}_2 = \vec{0}_2 + \vec{0}_1 = \vec{0}_1 = \vec{0}_2$$

□

6. Proof $0\vec{x} = \vec{0}, \forall \vec{x} \in V$

$$0\vec{x} = (0 + 0)\vec{x} = 0\vec{x} + 0\vec{x} \implies \vec{0} = 0\vec{x}$$

1.2 Subspaces and Spans

Definition – Subspace

Let V be a vector space over $\mathbb{F}$ and $W \subseteq V$. We say **subspace** of V if W is also a vector space over $\mathbb{F}$ with same addition and scalar multiplication as for V

Info – Subspace Test

Let V be a vector space over $\mathbb{F}$. Let $W \subseteq V$.

Then W is a subspace of V if and only if:

1. $\vec{0} \in W$
2. W is closed under addition (if $\vec{x}, \vec{y} \in W, \vec{x} + \vec{y} \in W$)
3. W is closed under scalar multiplication (if $\vec{x} \in W, c \in \mathbb{F}, c\vec{x} \in W$)

Example:

1. Let $W = \{p(x) \in P_0(\mathbb{R}) : p'(0) = 0\}$

W is a vector space with usual $+$ and $\cdot$ of polynomials

Most of the axioms are obvious for all polynomials.

We need to check axioms that are less intuitive:

1. Is the zero vector in W ?
2. Is W closed under addition?
3. Is W closed under scalar multiplication?

Proof

Let $V \in P_n(\mathbb{R})$

1. Is the zero vector in W ?

$$0 + 0x + \dots + 0x^n = z(\vec{x})$$

Clearly, $z'(0) = 0$ So $z(\vec{x}) \in W$

2. Is W closed under addition?

Let $p(x), q(x) \in W$

$$(p + q)'(0) = p'(0) + q'(0) = 0 \in W$$

3. Is W closed under scalar multiplication?

Let $p(x) \in W, s \in \mathbb{R}$.

$$(sp)'(0) = s(p'(0)) = s0 = 0 \in W$$

□

2. Let V be any vector space.

$W = V$ is a subspace. $W \setminus \{\vec{0}\}$ is a subspace.

3. $V = \mathbb{F}^n$

The origin is a subspace.

The lines pass through origin are subspace.

The planes pass through origin are subspace.

4. $V = P_3(\mathbb{F}), W = \{a + bx + cx^2 + dx^3 : b \geq 0\}$

W is not a subspace: not closed under scalar multiplication.

$$2x \in W, -1(2x) \notin W$$

4. $V = P_n(\mathbb{F}), W = \{p(x) \in V : p''(1) + p'(2) = 2\}$

W is not a subspace since $\vec{0} \notin W$

Note:

- A tip is that **non-homogeneous and equations** are usually **not** a subspace
- **Linear homogeneous** equations are usually a subspace

Definition – Span

Let V be a vector space. Let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in V$.

Then the **span** of $\vec{v}_1, \dots, \vec{v}_k$ is the set

$$W = \text{span}\{\vec{v}_1, \dots, \vec{v}_k\} = \{a_1\vec{v}_1 + \dots + a_k\vec{v}_k : a_i \in \mathbb{F}\}$$

Theorem

Let V be a vector space, let $\vec{v}_1, \dots, \vec{v}_k$ then $W = \text{span}\{\vec{v}_1, \dots, \vec{v}_k\}$ is a subspace of V

Note: the example (1) below gives a second proof that $\{A \in M_{2 \times 2} : \text{tr}(A) = 0\}$ is a subspace of $M_{2 \times 2}(\mathbb{F})$

Example:

1. $V = M_{n \times n}(\mathbb{F}), W = \{A \in M_{n \times n} : \text{trace}(A) = 0\}$

Recall that $\text{trace}(A)$ is the sum of the diagonal (i.e. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \text{trace}(A) = a + d$)

Claim W is a subspace of V

Recall that

- $\text{trace}(A + B) = \text{trace}(A) + \text{trace}(B)$
- $\text{trace}(cA) = c \text{trace}(A), c \in \mathbb{R}$

Proof

1. $\vec{0} = \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \dots & 0 \end{bmatrix}, \text{trace}(\vec{0}) = 0 + \dots + 0 = 0 \implies \vec{0} \in W$

2. Let $A, B \in W, c \in \mathbb{F}$.

$$\text{trace}(cA + B) = \text{trace}(cA) + \text{trace}(B) = c \text{trace}(A) + \text{trace}(B) = 0 \text{ since } A, B \in W$$

Thus $cA + B \in W$, closed by addition and scalar multiplication.

□

Take $W = \{A \in M_{2 \times 2}(\mathbb{F}) : \text{trace}(A) = 0\}$

For an arbitrary $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in W$

$$\iff \text{trace}(A) = 0 \iff a + d = 0 \iff d = -a \iff A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix} (a, b, c \in \mathbb{F})$$

$$\text{So } W = \left\{ \begin{bmatrix} a & b \\ c & -a \end{bmatrix} : a, b, c \in \mathbb{F} \right\} = \left\{ \begin{bmatrix} a & 0 \\ 0 & -a \end{bmatrix} + \begin{bmatrix} 0 & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ c & 0 \end{bmatrix} : a, b, c \in \mathbb{F} \right\}$$

$$\implies W = \text{span} \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

We can conclude that most of the time **imposing condition decreases degrees of freedom**

Note: differentiation and equation are **linear** operations

2. $V = P_2(\mathbb{R}), W = \{p(x) \in V : p'(1) = p(-1)\}$

For W , an arbitrary $p(x) = a + bx + cx^2 \in W$

$$p'(x) = b + 2cx \implies p'(1) = b + 2c, p(-1) = a - b + c$$

$$\text{Then } p'(1) = p(-1) \iff b + 2c = a - b + c \iff a - 2b + c = 0$$

$$\iff a = 2b + c \iff p(x) = a + bx + cx^2 = (2b + c) + bx + cx^2$$

$$\text{Now } W = \{(2b + c) + bx + cx^2 : b, c \in \mathbb{F}\} = \{b(2 + x) + c(1 + x^2) : b, c \in \mathbb{F}\}$$

$$\text{So } W = \text{span}\{2 + x, 1 + x^2\}$$

Note:

1. Writing a subspace W in the form $W = \text{span}\{\vec{v}_1, \dots, \vec{v}_k\}$ is that it is easier to write down vectors in W

2. Having $W = \text{span}\{\vec{v}_1, \dots, \vec{v}_k\}$ allows us to detect the $\dim W$. Here is a catch, some spanning set are “better” than others

(i.e. $W = \text{span}\{2 + x, 1 + x^2\}$, instead someone came up with

$$T = \text{span}\{2 + x, 1 + x^2, 4 + x + 2x^2\} = \text{span}\{p(x), q(x), r(x)\})$$

There is some linear relations between vectors in T :

$$r(x) = p(x) + 2q(x), p(x) = r(x) - 2q(x), q(x) = \frac{1}{2}r(x) - \frac{1}{2}p(x)$$

$$\implies p(x) + 2q(x) + (-1)r(x) = \vec{0}$$

1.3 Linear Independence

Definition – Linear Independence

Let $\vec{v}_1, \dots, \vec{v}_k \in V$.

1. $\vec{v}_1, \dots, \vec{v}_k$ are **linearly dependent** if $\vec{0} = c_1\vec{v}_1 + \dots + c_k\vec{v}_k, c_i \neq 0$
2. $\vec{v}_1, \dots, \vec{v}_k$ are **linearly independent** if they are not linearly dependent.

Theorem:

1. $\vec{v}_1, \dots, \vec{v}_k$ are **linearly dependent** if and only if some v_i is a linear combination of the others.
2. $\vec{v}_1, \dots, \vec{v}_k$ are **linearly independent** if and only if $c_1\vec{v}_1 + \dots + c_k\vec{v}_k = \vec{0} \implies c_1 = \dots = c_k = 0$

Examples:

1. $2 + x, 1 + x^2$ are linearly independent in $P_2(\mathbb{F})$ from the previous example.

Two vectors $\vec{v}_1, \vec{v}_2$ are linearly dependent if and only if they are a scalar multiple of the others. So they are not linearly dependent.

2. Let $S = \{1 - x, 1 + x, 1 - x^2, 1 + x^2\} \subseteq P_2(\mathbb{R})$, is S linearly independent?

Informally, $\dim P_2(\mathbb{R}) = 3$, but S contains 4. Thus there are too many to be linearly independent.

ANS:

Consider the equation $c_1(1 - x) + c_2(1 + x) + c_3(1 - x^2) + c_4(1 + x^2) = \vec{0}$

We want to solve c_1, c_2, c_3, c_4 and hope to get $c_1 = c_2 = c_3 = c_4 = 0$ is the only solution.

$$(c_1 + c_2 + c_3 + c_4) + (-c_1 + c_2)x + (-c_3 + c_4)x^2 = 0 + 0x + 0x^2$$

$$\begin{cases} c_1 + c_2 + c_3 + c_4 = 0 \\ -c_1 + c_2 = 0 \\ -c_3 + c_4 = 0 \end{cases} \implies \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \end{array} \right]$$

There will be a nontrivial solution if and only if $\text{null}(A) \neq 0$

Recall the Rank-Nullity Theorem: $n = \text{rank}(A) + \text{null}(A)$

Notice that $\text{rank}(A) \leq 3$, therefore, $\text{null}(A) = 1 \neq 0$

Since the $\text{null}(A) \geq 0$, there exists nontrivial solution. Thus S is linearly dependent.

3. $T = \{1 + x, 1 + x^2\}$ Does T span $P_2(\mathbb{R})$. That is, is $\text{span}(T) = P_2(\mathbb{R})$?

We expect the answer to be no. There are only 2 degrees of freedom for T . But, $\dim P_2(\mathbb{R}) = 3$. That is T is lacking one element to span a 3-dimensional space

To show that $\text{span } T \neq P_2(\mathbb{R})$. We need to show $\text{span } T \subseteq P_2(\mathbb{R})$ and $P_2(\mathbb{R}) \not\subseteq \text{span } T$.

Take $p(x) = a + bx + cx^2 \in P_2(\mathbb{R})$.

$$p(x) \in \text{span } T \iff p(x) = c_1(1 + x) + c_2(1 + x^2), \quad c_1, c_2 \in \mathbb{R}$$

$$a + bx + cx^2 = (c_1 + c_2) + c_1x + c_2x^2$$

$$\begin{cases} c_1 + c_2 = a \\ c_1 = b \\ c_2 = c \end{cases} \implies \left[\begin{array}{cc|c} 1 & 1 & a \\ 1 & 0 & b \\ 0 & 1 & c \end{array} \right]$$

We want the system to be consistent for all choices.

This is the case if and only if $\text{rank } A = n = 3$. But notice $\text{rank } A \leq 2 < 3$

Thus there exists some a, b, c s.t. system is inconsistent.

$p(x) \notin T$, so $p_2(\mathbb{R}) \not\subseteq \text{span } T$

Definition – Basis

Let V be a vector space. A **basis** for V is a set $B \subseteq V$ satisfy:

1. $B \subseteq V$
2. $\text{span } B = V$
3. B is linearly independent

Note: In general $E_{ij} \in M_{m \times n}(\mathbb{F})$ has the ij -th entry be 1.

Examples:

1. $S' = \{1 + x, 1 - x, 1 - x^2\}$ is a basis for W
2. $\{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n\}$ is the standard basis for $\mathbb{F}^n$
3. $\{1, x, x^2, \dots, x^n\}$ is a standard basis for $P_n(\mathbb{F})$
4. $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} = \{E_{11}, E_{12}, E_{21}, E_{22}\}$ is a standard basis for $M_{2 \times 2}(\mathbb{F})$
5. $\{E_{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$ is the standard basis for any $M_{m \times n}(\mathbb{F})$
6. In general, vector space V does not have a **standard basis**
7. Let W be the plane in $\mathbb{R}^3$ with equation $x + y + z = 0$

W is a 2-dimensional subspace in a 3-dimensional space

Then $B = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\} \subseteq W$ and B is linearly independent

Let $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \in W$, then $x + y + z = 0 \implies y = -x - z$

$$\implies \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ -x-z \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \in \text{span } B$$

Thus B is a basis for W . There are many other possible choices for basis for W . Any two linearly independent vectors in W satisfy to be a basis.

8. $V = \{A \in M_{2 \times 2}(\mathbb{F}) : \text{tr } A = 0\} \in M_{2 \times 2}(\mathbb{F})$

Find a basis for V

V is 4-dimensional space, we suspect the basis are less than 4-dimensional due to presence of imposed linear condition.

$$B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}.$$

Since B is linearly independent, so it is a basis for V

9. $W = \left\{ A \in M_{2 \times 2}(\mathbb{F}) : A \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} A \right\}$

This is a subspace of $M_{2 \times 2}(\mathbb{F})$

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $A \in W \iff \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \iff \begin{bmatrix} a & -b \\ c & -d \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & -d \end{bmatrix}$

$$\implies -b = 0, -c = c \iff b = 0, c = 0$$

$$A = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore $B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$, and it is linearly independent, therefore, it is a basis.

10. $V = \{\vec{0}\}$, zero vector space.

By convention, $B = \emptyset$ is a basis for V .

$\emptyset$ is linearly independent, $\text{span } \emptyset = \{\vec{0}\}$

QUESTION: Does every vector space have a basis?

ANSWER: Yes but for finite-dimensional vector space for this course.

(i.e. This is infinity-dimensional: $V = F(\mathbb{R}, \mathbb{R}) = \{\text{functions}, f : \mathbb{R} \rightarrow \mathbb{R}\}$)

Definition – Finite-Dimensional

A vector space is **finite-dimensional** if it has a finite spanning set: $V = \text{span}\{\vec{v}_1, \dots, \vec{v}_k\}$

$M_{m \times n}(\mathbb{F})$ and $P_n(\mathbb{F})$ are finite-dimensional spaces.

Spaces of functions: $f : \mathbb{R} \rightarrow \mathbb{R}$ are infinity-dimensional spaces.

Definition – Vector Space Basis

Let V be a finite-dimensional vector space.

1. V has a basis
2. Any two bases of V have the same size. (In other words, if B and C are bases for V , then $|B| = |C|$)
3. Every subspace of V is finite-dimensional

Proof

1. V has a basis

By definition, V has a finite spanning set: $S = \{\vec{v}_1, \dots, \vec{v}_k\}$

If S is linearly independent, then S is a basis.

If not, say $\vec{v}_k$ is a linear combination of others, then remove $\vec{v}_k$ and call it $S' = \{\vec{v}_1, \dots, \vec{v}_{k-1}\}$

Now S' spans V

2. Any two bases of V have the same size

Let B and C be two bases for V ,

if $|B| > |C|$, $|B|$ is not linearly independent.

if $|B| < |C|$, B does not span V .

$\implies |B| = |C|$

Tip – Proposition

Let V be a vector space with basis B of size n

Let S be a finite subset of V

- if $|S| > n$, then S is linearly independent
- if $|S| < n$, then S cannot span V

Definition – Dimension

Let V be finite-dimensional. Then the **dimension** of V $\dim V$ is the size of any basis for V

Remark: By the theorem, $\dim V$ is well-defined

Example:

V	Basis	$\dim V$
$\{\vec{0}\}$	$\emptyset$	0
$\mathbb{F}^n$	$\{\vec{e}_1, \dots, \vec{e}_n\}$	n
$M_{m \times n}(\mathbb{F})$	$\{E_{11}, E_{1i}, \dots, E_{ij}\}$	mn
$P_n(\mathbb{F})$	$\{1, x, \dots, x^n\}$	$n + 1$
Plane in $\mathbb{R}^n$ through $\vec{0}$	Any 2 LI vectors in plane	2
$\{A : M_{2 \times 2}(\mathbb{F}) : \text{tr} A = 0\}$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$	3
$\text{span}\{x, x + x^2, x - x^2\} \subseteq P_{10}(\mathbb{R})$	$\{x, x + x^2\}$	2
$\{p(x) \in P_3(\mathbb{R}) : p(1) = 0, p(-1) = 0\}$	$\{x^2 - 1, x^3 - x\}$	2

Note: V in the last row is a subset of $P_3(\mathbb{R})$, not necessarily be 4 dimensional

We need to find a spanning set for $\{p(x) \in P_3(\mathbb{R}) : p(1) = 0, p(-1) = 0\}$

$$p(x) = a + bx + cx^2 + dx^3 \in V \iff \begin{cases} p(1)=0 \\ p(-1)=0 \end{cases} \iff \begin{cases} a+b+c+d=0 \\ a-b+c-d=0 \end{cases}$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array} \right] \implies a = -c, b = -d \implies p(x) = -c + -dx + cx^2 + dx^3$$

$$\implies c(x^2 - 1) + d(x^3 - x) \implies V = \text{span}\{x^2 - 1, x^3 - x\}$$

Notice that $B = \{x^2 - 1, x^3 - x\}$ is a spanning set for V and is linearly independent, thus it is a basis for V

Definition – Dimension and Vector Space

Let V be a finite-dimensional vector space. Let $W \subseteq V$ be a subspace.

1. W is finite-dimensional
2. $\dim W \leq \dim V$
3. $\dim W = \dim V \iff W = V$

Proof

2. $\dim W \leq \dim V$

Let B be a basis for W ($|B| = \dim W$).

Since $B \subseteq W \subseteq V$, we have $|B|$ linearly independent vectors in V .

But $|\text{Linearly independent}| \leq \dim$, so $|B| \leq \dim V$.

Thus $\dim W \leq \dim V$.

3. $\dim W = \dim V \iff W = V$

$\implies$

We have that $\dim W = \dim V$, then if B is a basis for W , then it is also a basis for V . Since B is linearly independent having $\dim V$. Thus $V = \text{span}\{B\} = W$

Conversely also applies.

1.4 Coordinates

In $\mathbb{R}^3$, let $S = \left\{ \vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

Then any $\vec{x} \in \mathbb{R}^3$ is a linear combination of $\vec{e}_1, \vec{e}_2, \vec{e}_3$

In $\mathbb{R}^2$, let $B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. This is a basis.

Consider $\vec{v} = \begin{bmatrix} 3 \\ 5 \end{bmatrix} \in \mathbb{R}^2$

Then $\vec{v} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

 Definition – Coordinates

Let $B = \{\vec{b}_1, \dots, \vec{b}_n\}$ be an ordered basis for V ,

Then the B – coordinates of $\vec{v} \in V$ are the coefficients of $\vec{v}$ when we express it as a linear combination of V

$$\vec{v} = a_1 \vec{b}_1 + \dots + a_n \vec{b}_n$$

The B – coordinates vector of $\vec{v}$ is $[\vec{v}]_B = \begin{bmatrix} a_1 \\ \dots \\ a_n \end{bmatrix}$

Examples:

Let $V = P_3(\mathbb{R})$, $B = \{1, x, x^2, x^3\}$

Then $p(x) = 1 - x^3 = 1 + 0x + 0x^2 - x^3$